

The Future of the Sun and the Fate of the Earth: A Calibrated Stellar Model Coupled to an N -body Integration of the Surviving Planets

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ABSTRACT

We compute the evolution of the Sun from the present day through the red-giant branch (RGB), helium flash, asymptotic giant branch (AGB), and into the white-dwarf (WD) cooling track using a solar-calibrated MESA model, and couple the resulting mass-loss and radius histories to a direct N -body integration of the surviving planets that includes general relativity and the convective equilibrium tide. The calibrated model reproduces the present-day Sun ($L = 0.997 L_{\odot}$, $R = 1.000 R_{\odot}$, $T_{\text{eff}} = 5767$ K) and predicts a moist-greenhouse threshold for Earth in 1.15 Gyr and a runaway greenhouse in 3.7 Gyr. At the RGB tip (7.39 Gyr hence) the Sun reaches $R = 202 R_{\odot} = 0.94$ AU at $2760 L_{\odot}$, having shed 24% of its mass; Mercury and Venus are engulfed at pericenter contact 4.7 and 0.7 Myr before the tip, respectively, while the Earth survives, its orbit carried outward to ≈ 1.25 AU by mass loss. We identify a dynamical consequence that is absent from prior single-planet engulfment studies and from uniform-mass-loss N -body studies: because tidal drag suppresses Earth’s orbital expansion near the tip while Mars expands adiabatically, the Earth–Mars period ratio sweeps through the 7:3 and 9:4 mean-motion resonances, pumping both eccentricities until the orbits cross. In our fiducial realization Mars is ejected and the Earth is left on an eccentric orbit ($a \approx 1.67$ AU, $e \approx 0.17$) about a $0.53 M_{\odot}$ carbon–oxygen white dwarf. The instability is reproduced by two independent integrators. We discuss the sensitivity of this *tidally-differential* destabilization channel to the RGB-tip radius, the wind efficiency, and the tidal prescription, and outline the ensemble calculation needed to establish its branching ratios. All inputs, code, and a scrub-controllable visualization are publicly available.

Keywords: Solar evolution — Stellar mass loss — Planetary dynamics — Tidal interaction — White dwarf stars

1. INTRODUCTION

The long-term future of the Sun is a classic problem (Sackmann et al. 1993; Rybicki & Denis 2001; Schröder & Connon Smith 2008). As the Sun exhausts its central hydrogen, ascends the RGB, and ejects its envelope, its luminosity and radius vary by orders of magnitude, and roughly half of its present mass is returned to the interstellar medium. The fate of the Earth in this endgame has been debated for decades. Sackmann et al. (1993) and Rybicki & Denis (2001) established the timeline of the Sun’s brightening and the associated loss of Earth’s habitability, while Schröder & Connon Smith (2008) performed a careful calculation of the RGB-tip radius and the competition between wind-driven orbital expansion and tidal drag, concluding that the Earth is engulfed just short of the tip.

Two threads of prior work bracket the problem but do not meet. On one side, single-planet engulfment calculations (Rasio et al. 1996; Villaver & Livio 2009; Mustill

& Villaver 2012; Schröder & Connon Smith 2008) treat the tidal in-spiral of one planet into an evolving giant with care, but by construction they say nothing about the *multi-planet* dynamics of the survivors. On the other side, the classic post-main-sequence N -body studies (Duncan & Lissauer 1998; Veras 2016) integrate the full planetary system through mass loss and find the surviving planets dynamically stable — but they adopt *uniform* (adiabatic) mass loss, which preserves period ratios and Hill spacings exactly, and they omit stellar tides. The regime in which tides act *differentially* on the inner survivors, breaking the uniform expansion that underpins the stability argument, has not been examined for the solar system.

Here we close that gap. We evolve a solar-calibrated MESA model (Paxton et al. 2011, 2019) from the pre-main sequence to the WD stage, and drive a direct N -body integration of the eight planets with the time-dependent solar mass and radius, general-relativistic precession, and the convective equilibrium tide. Section 2 describes

the stellar model, the tidal prescription, and the integration strategy. Section 3 presents the climate timeline, the structural evolution, the engulfment of Mercury and Venus, the survival of the Earth, and the tidally-driven Earth–Mars instability. Section 4 compares our results to prior work and enumerates the model sensitivities. Section 5 concludes.

2. METHODS

2.1. Solar-calibrated stellar model

We use MESA release r26.04.1 with the MESA SDK. The initial helium abundance Y_0 , metallicity Z_0 (parameterized by $[\text{Fe}/\text{H}]$), mixing length α_{MLT} , and exponential overshoot efficiency f_{ov} are calibrated with MESA’s simplex framework to match, at the solar age of 4.61 Gyr, the solar luminosity and radius, the surface metal-to-hydrogen ratio Z/X , the surface helium abundance, the depth of the convective envelope, and the helioseismic sound-speed profile. Element diffusion is included and gravitational settling is followed. The best-fit parameters are $Y_0 = 0.2772$, $Z_0 = 0.0203$ ($[\text{Fe}/\text{H}] = +0.10$), $\alpha_{\text{MLT}} = 1.945$, and $f_{\text{ov}} = 0.0314$, with a combined $\chi^2 = 3.54$; the resulting present-day model has $L = 0.997 L_\odot$, $R = 1.000 R_\odot$, $T_{\text{eff}} = 5767$ K, i.e. a luminosity residual of 0.3% — roughly three times the amplitude of the 11-year solar cycle in total irradiance.

The calibrated model is evolved from the pre-main sequence through the WD stage in six segments (pre-MS $\rightarrow$ terminal-age main sequence $\rightarrow$ RGB tip $\rightarrow$ core helium exhaustion $\rightarrow$ end of the AGB $\rightarrow$ WD), with mass loss on the giant branches following the Reimers (1975) prescription on the RGB and a Blöcker (1995) superwind on the AGB. History output is written at every timestep to resolve the rapid tip and thermal-pulse phases, providing continuous $M_\star(t)$, $R_\star(t)$, $L_\star(t)$, and interior structure for the dynamical coupling.

2.2. The convective equilibrium tide

Near the RGB tip the Sun develops a deep convective envelope, and the equilibrium tide raised by an inner planet is damped by turbulent convection, driving orbital decay (Zahn 1977, 1989; Verbunt & Phinney 1995). We adopt the standard form for the tidally-driven decay of the semi-major axis a of a planet of mass m (mass ratio $q \equiv m/M_\star$) orbiting a star of mass $M_\star$, envelope mass M_{env} , and radius $R_\star$:

$$\left(\frac{1}{a} \frac{da}{dt}\right)_{\text{tide}} = -\frac{4}{7} f \frac{M_{\text{env}}}{M_\star} \frac{q(1+q)}{\tau_{\text{conv}}} \left(\frac{R_\star}{a}\right)^8, \quad (1)$$

where the convective turnover time is $\tau_{\text{conv}} = (M_{\text{env}} R_\star^2 / L_\star)^{1/3}$ and

$$f = \min \left[1, \left(\frac{P_{\text{orb}}}{2\tau_{\text{conv}}} \right)^2 \right] \quad (2)$$

is the Goldreich & Nicholson (1977) reduction factor that suppresses the turbulent viscosity when the tidal forcing period is shorter than the convective turnover time. The steep $(R_\star/a)^8$ dependence makes the tide negligible until $R_\star$ climbs to within a factor of a few of the orbit, then switches on abruptly; it also renders the innermost surviving planet overwhelmingly dominant. Orbital expansion from mass loss is not imposed as a separate term: the stellar mass is updated directly in the N -body integration, and the adiabatic scaling $a \propto M_\star^{-1}$ emerges from the dynamics. We validate this by comparison with the analytic secular solution (Section 2.4).

2.3. N -body integration

We integrate the Sun and eight planets with REBOUND (Rein & Liu 2012) and REBOUNDx (Tamayo et al. 2020). General-relativistic apsidal precession is included with the `gr_potential` force, which is position-dependent and therefore symplectic-safe. The tide of Equation (1) is applied as a per-planet semi-major-axis decay operator. On the quiet RGB climb we use the WHFast integrator (Rein & Tamayo 2015) with a timestep of $P_{\text{Mercury}}/10$; the stellar mass, radius, and tidal rates are updated each chunk from the MESA history, with chunk lengths adapted so that $|\Delta M_\star|/M_\star < 10^{-4}$ and $|\Delta R_\star|/R_\star < 1\%$. A planet is removed when its pericenter first contacts the photosphere, $a(1-e) < R_\star$, at which point gas drag renders the remaining in-spiral effectively instantaneous.

Because WHFast cannot integrate close encounters, once the Earth–Mars orbits begin to cross (Section 3.4) we re-integrate the tail with the hybrid symplectic integrator MERCURIUS (Rein et al. 2019), restarted from the last pre-crossing snapshot with a timestep of $P_{\text{Earth}}/40$. To make the multi-Gyr integration tractable we begin the N -body run at 10.75 Gyr (present age +6.14 Gyr); at that epoch the Sun has lost $< 1\%$ of its mass and the tidal rates are $\sim 10^{-21} \text{ yr}^{-1}$, so the planetary orbits are indistinguishable from their present values, and every dynamically active phase is captured.

2.4. Validation

As an independent check we constructed a secular model in which each planet’s semi-major axis evolves under Equation (1) plus the mass-loss term $\dot{a}/a = -\dot{M}_\star/M_\star$, integrated with an adaptive ODE solver. Over the final 14 Myr before the RGB tip, the secular

model and the direct N -body integration agree on the Mercury and Venus engulfment times to within ~ 10 kyr across a 1.2 Gyr integration, and on the surviving orbits to a few parts in 10^4 ; the residuals are the genuine planet–planet perturbations that the secular model lacks. This confirms that the tidal prescription is correctly implemented and that the mass-loss expansion emerges properly from the N -body dynamics.

3. RESULTS

3.1. *The brightening Sun and the end of the biosphere*

The calibrated model brightens by $\sim 0.85\%$ per 100 Myr on the main sequence, steepening with time. The solar luminosity reaches $+6\%$ in 0.8 Gyr, $+10\%$ in 1.15 Gyr, and $+40\%$ in 3.7 Gyr. Coupled to standard climate results, this timeline implies the end of C3 photosynthesis and the decline of the complex biosphere within ~ 0.5 –1 Gyr as the carbonate–silicate cycle draws down CO_2 (Caldeira & Kasting 1992; O’Malley-James et al. 2013), a moist-greenhouse transition and the onset of ocean loss near $+1.15$ Gyr (Wolf & Toon 2015), and a full runaway greenhouse by $+3.7$ Gyr. The Earth is rendered uninhabitable at its surface long before the Sun leaves the main sequence.

3.2. *Structural evolution to the white-dwarf stage*

Central hydrogen is exhausted at $+4.43$ Gyr, after which the Sun crosses the subgiant branch and ascends the RGB. At the RGB tip, $+7.385$ Gyr from now (stellar age 11.95 Gyr), the model reaches $R_\star = 202 R_\odot = 0.94$ AU, $L_\star = 2760 L_\odot$, $T_{\text{eff}} = 2940$ K, with a degenerate helium core of $0.475 M_\odot$; 24% of the Sun’s mass has already been lost to the RGB wind. The degenerate core ignites in the helium flash; the Sun then settles onto the horizontal branch (the red clump), ascends the AGB, ejects its envelope in a thermal-pulse superwind, and crosses the top of the HR diagram at $T_{\text{eff}} > 26,000$ K as a planetary-nebula nucleus before cooling as a $0.53 M_\odot$ carbon–oxygen white dwarf.

3.3. *Engulfment of Mercury and Venus; survival of the Earth*

In the N -body integration, Mercury is engulfed at stellar age 11.9465 Gyr (4.7 Myr before the tip) when its pericenter, driven inward by the tide from $a = 0.41$ AU, contacts the 0.32 AU photosphere; its natural eccentricity $e \approx 0.2$ brings it to grief early. Venus follows at 11.9513 Gyr (0.7 Myr before the tip) at $a = 0.77$ AU against $R_\star = 0.77$ AU. Both times match the secular prediction to $\lesssim 2$ kyr. The Earth, in contrast, survives: solar mass loss expands its orbit to ≈ 1.25 AU by the time the photosphere peaks at 0.94 AU, and the tide

— reduced by the Goldreich & Nicholson (1977) factor because Earth’s orbital period is short compared to the turnover time near the tip — is too weak to overcome the expansion. This is the opposite verdict to Schröder & Connon Smith (2008), for reasons we discuss in Section 4.

3.4. *A tidally-differential Earth–Mars instability*

The novel result concerns the aftermath. Near the tip, the tide acts strongly on the Earth (suppressing its adiabatic expansion by $\sim 7\%$) but negligibly on Mars, which expands with the full mass loss. As a result the Earth–Mars period ratio, ≈ 2.45 today, sweeps downward through the 7:3 and 9:4 mean-motion resonances during and after the tip. The resonance crossings pump both eccentricities: within ~ 0.1 Gyr of the tip, $e_\oplus$ and e_{Mars} grow from ~ 0.02 to $\gtrsim 0.25$ –0.35, and the orbits cross. In our fiducial realization the ensuing close encounters eject Mars from the solar system and leave the Earth on an eccentric orbit, $a \approx 1.67$ AU, $e \approx 0.17$, about the newborn white dwarf. The instability is not a numerical artifact of the symplectic scheme: it appears identically in the WHFast integration (before it fails at orbit crossing) and in the encounter-capable MERCURIUS re-integration.

Physically, this is a channel that the two prior threads of work each miss by construction. Uniform mass loss (Duncan & Lissauer 1998) preserves the Earth–Mars period ratio and cannot drive a resonance sweep; single-planet tidal calculations (Schröder & Connon Smith 2008) do not track Mars. It is the *differential* action of the tide on two surviving planets — one deep enough in the potential to feel the tide, one not — that breaks the adiabatic invariance and destabilizes the pair.

4. DISCUSSION

4.1. *Comparison with Schröder & Connon Smith (2008)*

Our RGB-tip radius, 0.94 AU, is smaller than the 1.2 AU ($256 R_\odot$) of Schröder & Connon Smith (2008), and where they engulf the Earth we find it survives with a ~ 0.3 AU margin. The difference is the expected one: the tip radius depends on the adopted opacities, mixing length, and (through the core mass at ignition) the prior wind history, and the survival verdict is a near-cancellation between tip radius, wind efficiency, and tidal strength. Our tide includes the Goldreich & Nicholson (1977) reduction, which weakens Earth’s tide near the tip relative to a full-strength Zahn prescription. The Earth’s fate is therefore genuinely model-dependent, and we regard the survival/engulfment boundary as the central systematic uncertainty rather

than a settled outcome; the differential instability we identify operates in the branch where Earth survives.

4.2. Caveats and the required ensemble

Three caveats bound the Earth–Mars result. First, it is a *single realization* of a chaotic outcome; the Lyapunov time of the inner solar system is ~ 5 Myr, so the specific disposition (Mars ejected / Earth eccentric) is one draw, and the branching ratio between ejection, mutual collision, and a long-lived resonant chain requires an ensemble of integrations with perturbed initial conditions. Second, our tidal model damps semi-major axis but not eccentricity; including the eccentricity tide would modify the timing of the resonance crossings, though by less than the tip-radius uncertainty. Third, the final ~ 30 Myr of the integration relies on a modest extrapolation of the stellar track beyond the end of the MESA data. None of these affects the *qualitative* result — that differential tidal migration drives the surviving terrestrial planets through resonance — which is robust within the well-constrained portion of the integration.

4.3. Implications

If it survives an ensemble treatment, this channel changes the standard picture of the post-RGB solar system from “the survivors expand quietly and remain stable” to one in which the surviving terrestrial planet(s) are dynamically excited or lost. This is directly relevant to the interpretation of polluted white dwarfs (Veras 2016): a mechanism that excites and ejects surviving planets at the RGB tip is a natural way to stir small-body reservoirs and deliver material to the WD.

5. CONCLUSIONS

We have computed a solar-calibrated evolutionary model of the Sun from the present day into the white-dwarf era and coupled it to a direct N -body integration of the surviving planets including general relativity and the convective equilibrium tide. The main results are: (i) a moist-greenhouse end to Earth’s surface habitability in 1.15 Gyr; (ii) an RGB tip in 7.39 Gyr at $R_\star = 0.94$ AU, at which Mercury and Venus are engulfed but the Earth survives; and (iii) a previously unremarked *tidally-differential* destabilization of the Earth–Mars pair after the tip, which in our fiducial realization ejects Mars and leaves the Earth eccentric about a $0.53 M_\odot$ white dwarf. The instability is reproduced by two independent integrators and occupies the gap between single-planet engulfment studies and uniform-mass-loss N -body studies. An ensemble calculation with an eccentricity tide and extended stellar tracks is the natural next step.

DATA AVAILABILITY

All MESA inlists, the calibrated model configuration, the N -body driver, the validation and analysis code, and a scrub-controllable visualization are available at https://github.com/oklo/solar_evolution. Bulk data products (MESA histories and profiles, N -body archives) are regenerable from the provided workflow.

- 1 This work used MESA (Paxton et al. 2011, 2019), the
- 2 MESA SDK, and REBOUND/REBOUNDx (Rein & Liu 2012;
- 3 Tamayo et al. 2020).

Software: MESA (Paxton et al. 2011, 2019), REBOUND (Rein & Liu 2012), REBOUNDx (Tamayo et al. 2020), WHFast (Rein & Tamayo 2015), MERCURIUS (Rein et al. 2019), NumPy, SciPy, Matplotlib.

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